Formal Semantics

Concepts of Programming Languages

CAS CS 320

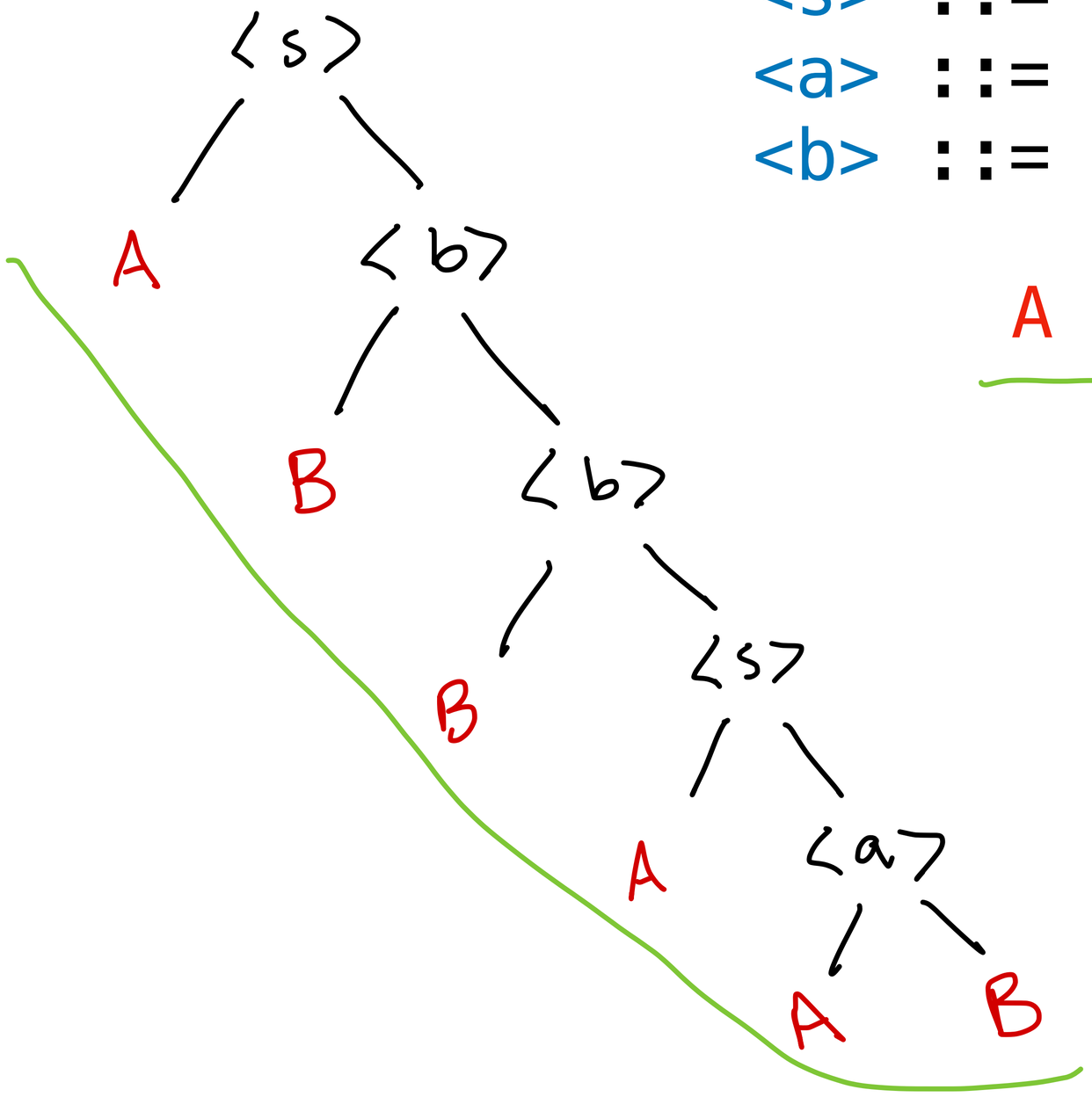
Practice Problem

```
<s> ::= A <a> | A <b>
<a> ::= A B
<b> ::= B <b> | B <s>
```

Is the following sentence recognized by the above grammar?

A B B A A B

Answer



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A B B A A B

Outline

- Discuss **formal semantics** in general
- Look at **small-step** and **big-step** semantics with some examples

Introduction

A Thought Experiment

```bash
x=3
function f () {
    x=2
}
f
echo $x
```

Bash

```python
x = 3
def f():
    x = 2
f()
print(x)
```

Python

```ocaml
let x = 3
let f () =
  let x = 2 in
  ()
let _ = f ()
let _ = print_int x
```

OCaml

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Usually we build intuitions by writing programs and reading manuals

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OCaml

3

Question. *How do we know what will happen when a program executes?*

Usually we build intuitions by writing programs and reading manuals

But many decisions about what it means to execute a program are arbitrary (or based on concerns like efficiency)

Meaning

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Syntax is interested in the *form* of a program

```
let x = 2 in
let y = 3 in
x + y
```

Meaning

Syntax is interested in the *form* of a program

Semantics is interested in the *meaning* of a program

```
let x = 2 in
let y = 3 in
x + y
```

↓

```
let y = 3 in
2 + y
```

↓

```
2 + 3
```

↓

```
5
```

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What is the meaning of meaning?

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Meaning

Syntax is interested in the *form* of a program

Semantics is interested in the *meaning* of a program

What is the meaning of meaning?

Formal semantics is the mathematical study of meaning

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Aside: Denotational vs. Operational Semantics

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Denotational semantics is interested in what a syntactic object "denotes" i.e. in interpreting programs as *objects in a mathematical space*

$$1 + 2 * 3 + 4 = 11$$
$$1 + 12 - 2 = 11$$

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Operational semantics is interested in how a programming language "operates" i.e. how a program *behaves* during execution

$$\begin{aligned} 1 + 2 * 3 + 4 &\longrightarrow 1 + 6 + 4 \\ &\longrightarrow 7 + 4 \\ &\longrightarrow 11 \end{aligned}$$

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Small-Step vs. Big-Step Semantics

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Small-step operational semantics is interested in *program transformation*, i.e., how a program transforms "one step at a time"

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Small-Step vs. Big-Step Semantics

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Big-step operational semantics is interested in *evaluation*, i.e., what is the value of the program once a program has finished evaluating

$$\dfrac{2 \Downarrow 2 \qquad \dfrac{3 \Downarrow 3 \qquad \dfrac{2 \Downarrow 2 \qquad 3 \Downarrow 3}{2 + 3 \Downarrow 5}}{\texttt{let y = 3 in 2 + y} \Downarrow 5}}{\textit{let x = 2 in let y = 3 in x + y} \Downarrow 5}$$

Small-Step vs. Big-Step Semantics

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Mini-projects

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Static vs. Dynamic Semantics

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Static semantics
refers to the meaning
given to a program
before it is evaluated

```
% ocaml silly.ml
File "./silly.ml", line 1, characters 8-9:
1 | let x = 2 +. 3.
            ^
Error: This expression has type int but an expression was
       expected of type
         float
  Hint: Did you mean `2.'?
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utop # let x = 2 + 3;;
val x : int = 5
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Type checking

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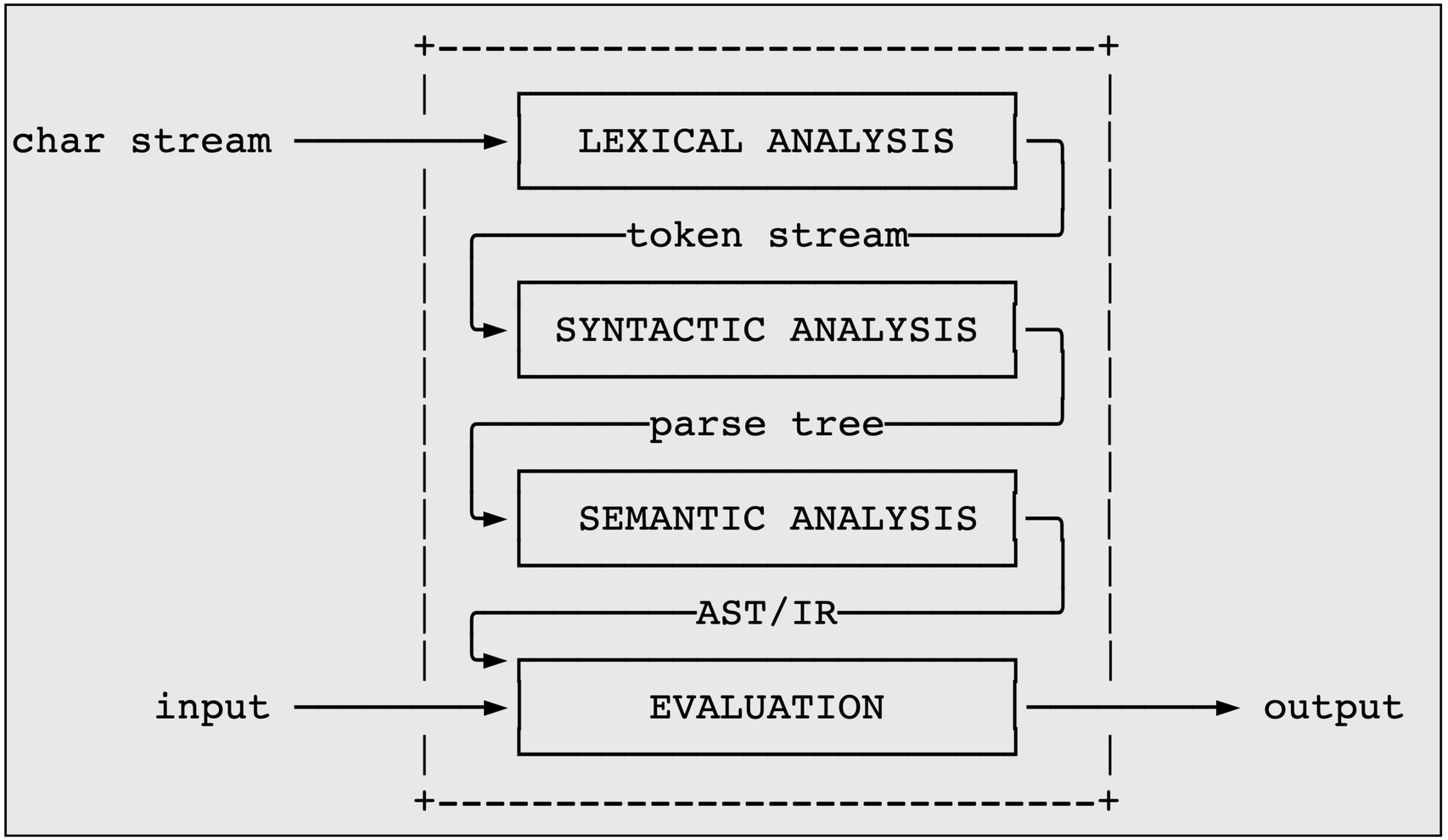
Recall: The Picture



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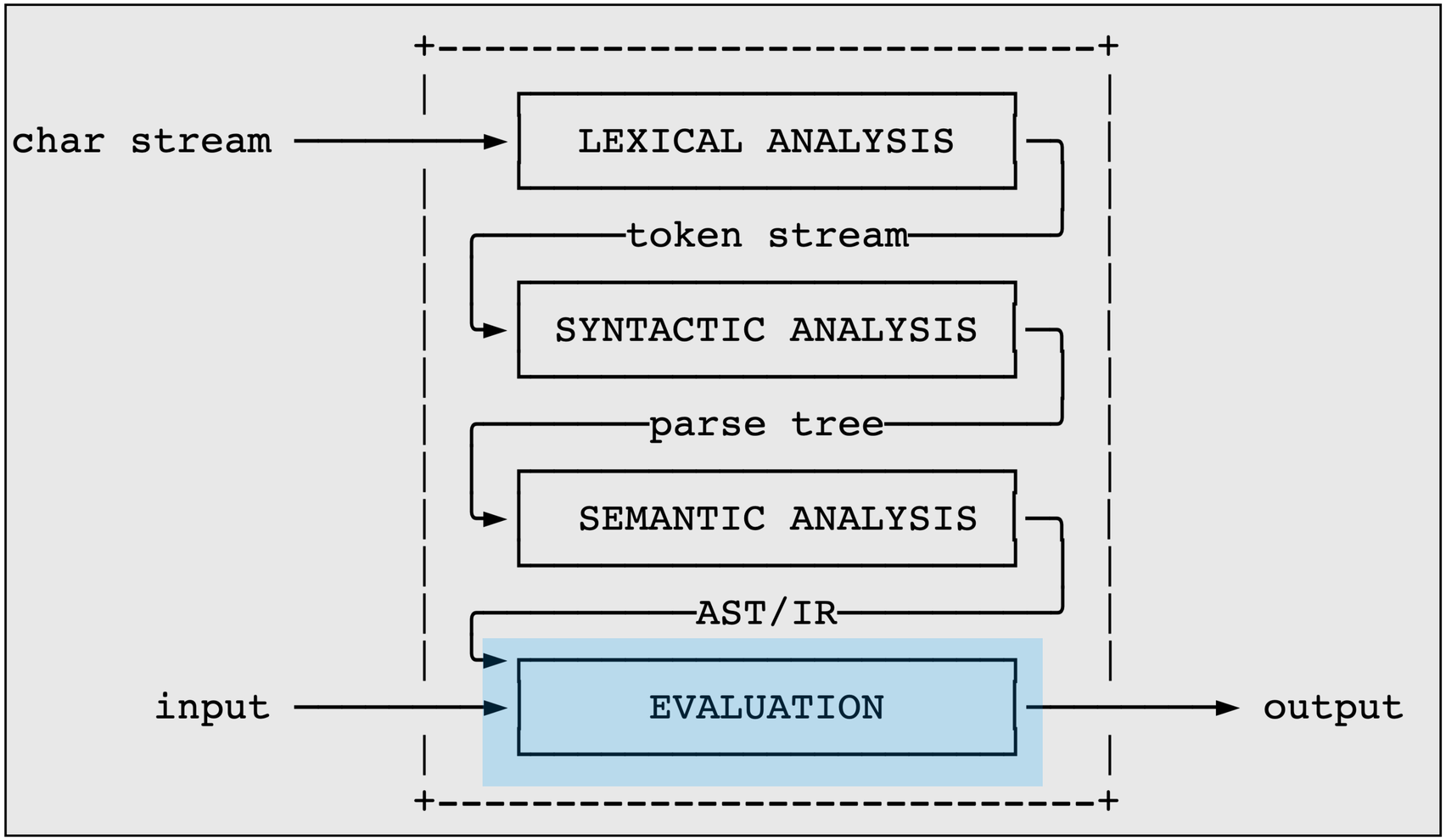

dynamic semantics (this week + next week)

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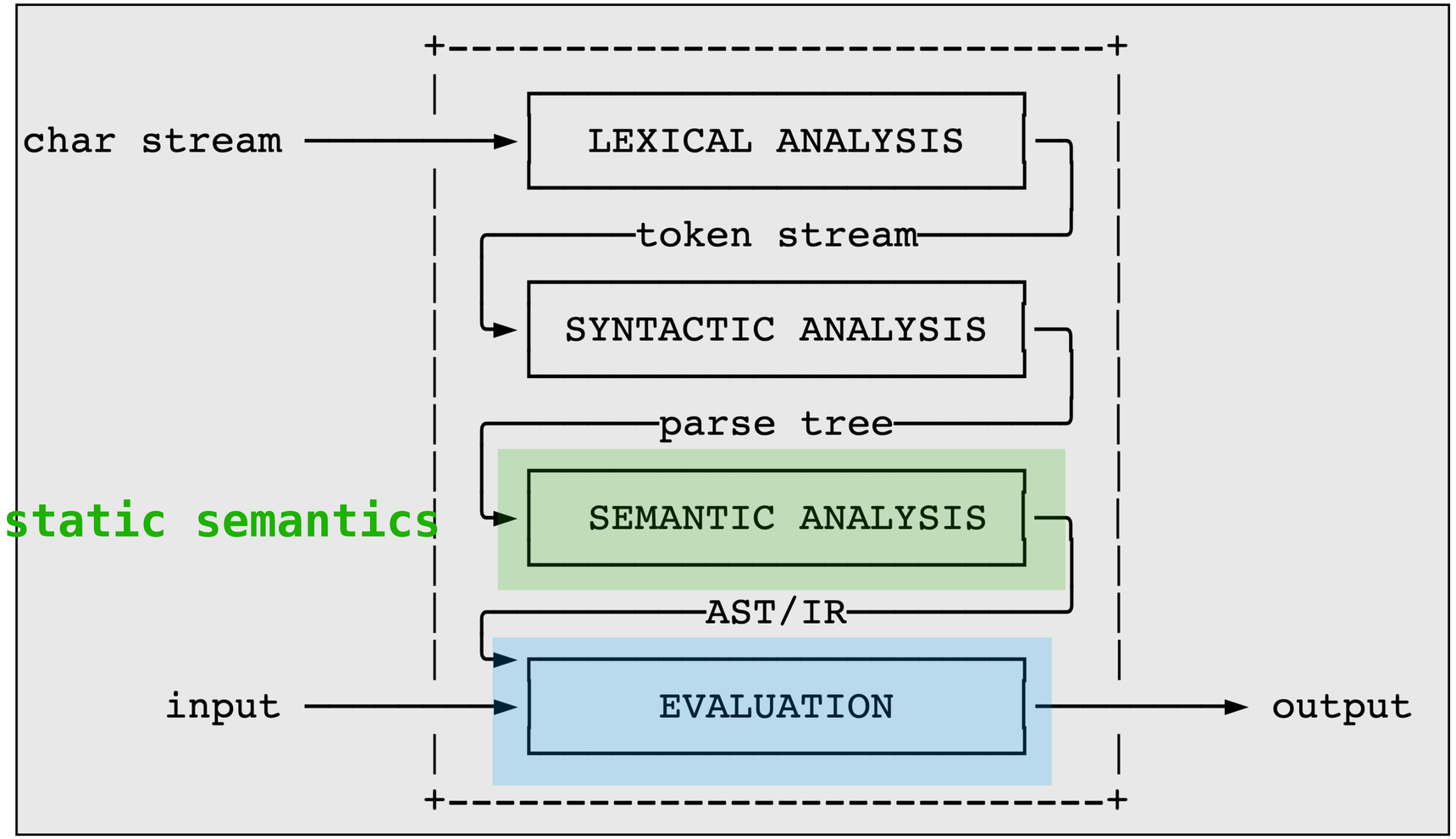

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Operational Semantics

Recall: Inference Rules

$$\frac{\overset{\text{premise}}{P_1} \quad \overset{\text{premise}}{P_2} \quad \cdots \quad \overset{\text{premise}}{P_k}}{\underset{\text{conclusion}}{C}}\ \overset{\text{name}}{\texttt{red-rule}}$$

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Then general form of a reduction rule has a collection of **premises** and a **conclusion**

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Then general form of a reduction rule has a collection of **premises** and a **conclusion**

There may be no premises, this is called an **axiom**

Example (Basic)

$$\frac{\overset{\text{premise}}{e_1 \longrightarrow e_1'}}{\underset{\text{conclusion}}{(\text{add } e_1\ e_2) \longrightarrow (\text{add } e_1'\ e_2)}}\ \text{add-left}$$

small-step sem. judmt.

```
<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
<op>   ::= add | sub  | eq
<bool> ::= true | false
<int>  ::= ...
```

Example Programs:

```
(add 2 3)
(add (add 2 3) 5)
(eq (add 2 3) (sub 7 2))
(add true 2)
```

Example (Basic)

$$\frac{\overbrace{e_1 \longrightarrow e_1'}^{\text{premise}}}{\underbrace{(\text{add } e_1\ e_2) \longrightarrow (\text{add } e_1'\ e_2)}_{\text{conclusion}}}\ \text{add-left}$$

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```
Example Programs:
(add 2 3)
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```

If e_1 *reduces to* e_1' *in one step, then* add e_1 e_2 *reduces to* add e_1' e_2 *in one step*

Example (Axiom)

$$\frac{n_1 \text{ is a number} \qquad n_2 \text{ is a number}}{(\texttt{add}\ n_1\ n_2) \longrightarrow n_1 + n_2}\ \texttt{add-ok}$$

If n_1 *and* n_2 *are numbers then* $(\texttt{add}\ n_1\ n_2)$ *reduces in one step to the number* $n_1 + n_2$

(In this case, the premises are side-conditions)

We'll come back to these examples...

Small-Step Semantic Judgment

$$(S, p) \longrightarrow (S', p')$$

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$2 + (3 \times 4)$

$2 + 12$

14

Small-step semantics formalizes a **"step by step"** computation which reduces a syntactic object until no reductions can be done

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In general, we define small-step semantics on a **configuration**, which is a program together with some stateful information

Small-Step Semantic Judgment

$$(S, \overset{\text{program}}{p}) \longrightarrow (S', \overset{\text{transformed program}}{p'})$$

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$$(\underbrace{S}_{\text{state}}, \overbrace{p}^{\text{program}}) \longrightarrow (\underbrace{S'}_{\text{transformed state}}, \overbrace{p'}^{\text{transformed program}})$$

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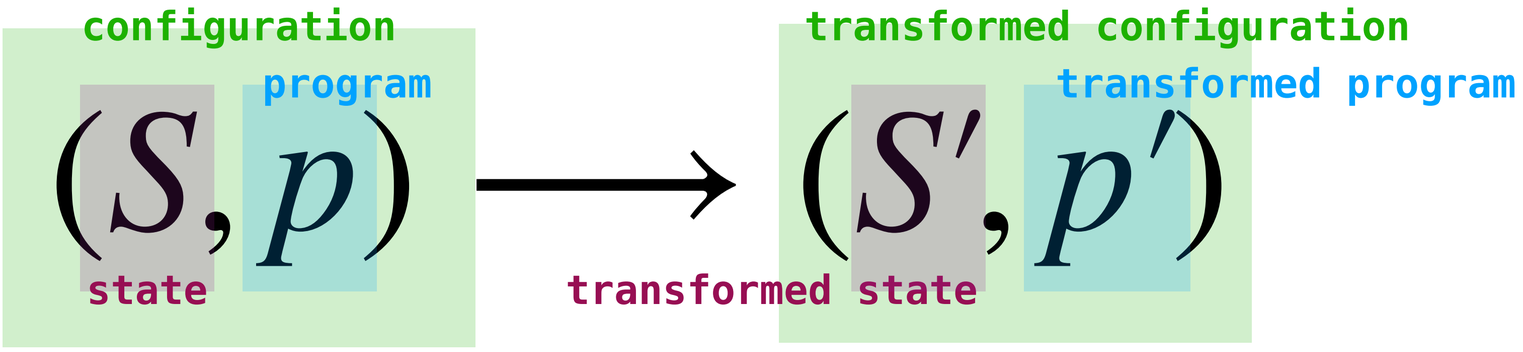

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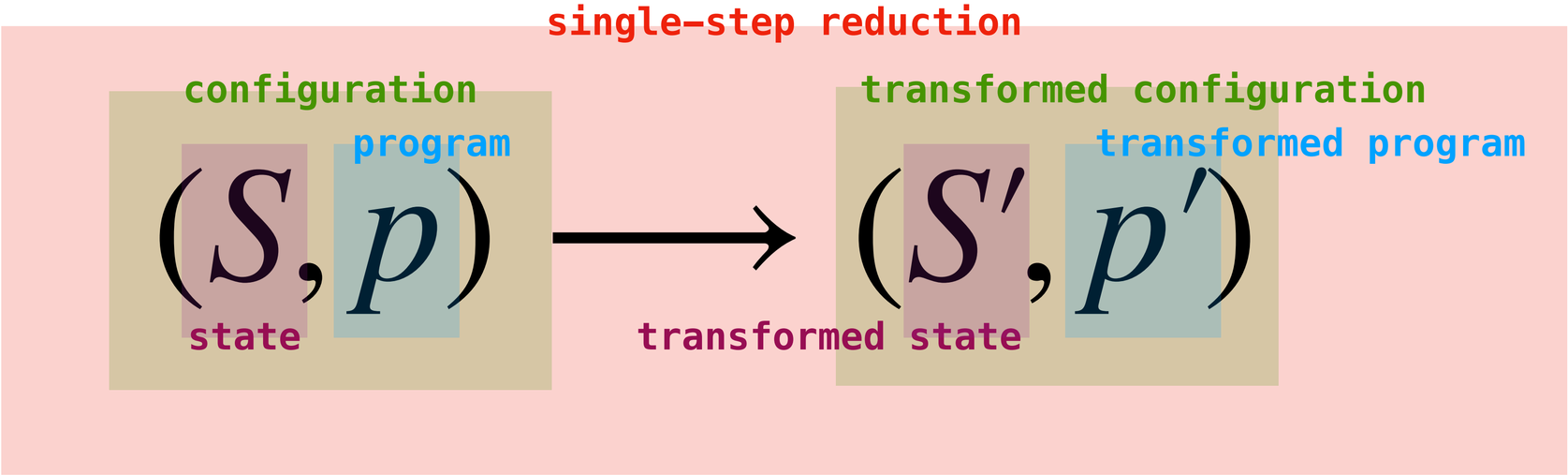

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Example: Arithmetic Expressions

$$(\underbrace{\varnothing}_{\texttt{state}}\ ,\ \overbrace{10 \times (2 + 3)}^{\texttt{program}}) \longrightarrow (\varnothing\ ,\ 10 \times 5) \longrightarrow (\varnothing\ ,\ 50)$$

<u>State:</u> none

<u>Program:</u> arithmetic expression

Example: (Fragment of) OCaml

program

(∅ , let x = 3 in if x > 10 then 4 else 5) ⟶ (∅ , if 3 > 10 then 4 else 5)

state

⟶ (∅ , if false then 4 else 5)

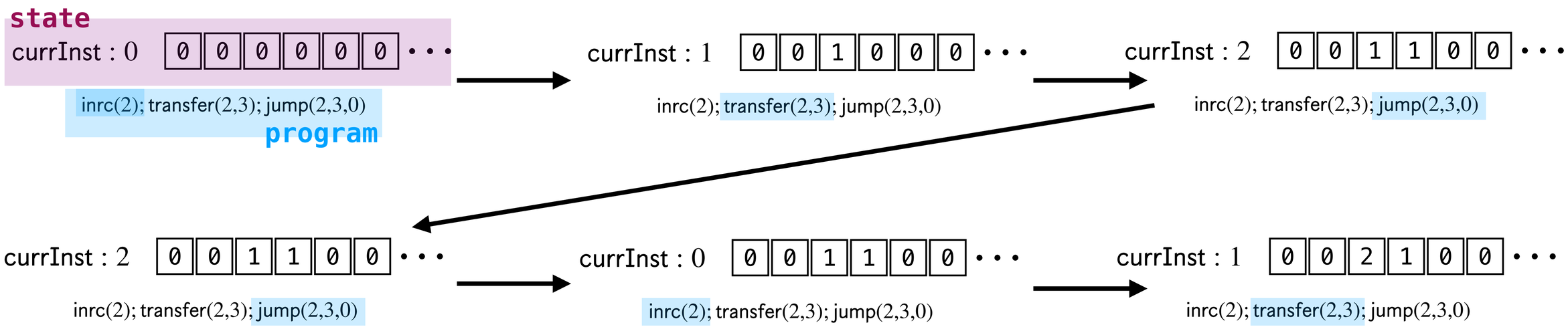
⟶ (∅ , 5)

<u>State:</u> none

<u>Program:</u> OCaml expression

For purely functional languages
there is no state

Example: Unlimited Register Machines



<u>State:</u> (current instruction pointer) + (collection of number registers)

<u>Program:</u> sequence of commands for updating registers values and current instruction

Example: Stack-Oriented Language

[ ]

state program

(∅ , push 2; push 3; add) ⟶

(2 :: ∅, push 3; add) ⟶

(3 :: 2 :: ∅, add) ⟶

(5 :: ∅, ϵ)

<u>State:</u> stack (i.e., list) of values

<u>Program:</u> sequence of commands for manipulating the stack

High-Level

$$(\underbrace{S}_{\text{state}}, \overbrace{p}^{\text{program}}) \xrightarrow[\text{transformed state}]{} (S', \overbrace{p'}^{\text{transformed program}})$$

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» What kind of **state** are we manipulating?

High-Level

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When we define the small-step semantics of PL, we need to define two things:

» What kind of **state** are we manipulating?

» What **rules** describe how to transform configurations?

Example

```
<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
<op>   ::= add | sub | eq
<bool> ::= true | false
<int>  ::= ...
```

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$$\frac{e_1 \longrightarrow e_1'}{(\texttt{add}\ e_1\ e_2) \longrightarrow (\texttt{add}\ e_1'\ e_2)}\ \texttt{add-left}$$

$$\frac{e_2 \longrightarrow e_2'}{(\texttt{add}\ e_1\ e_2) \longrightarrow (\texttt{add}\ e_1\ e_2')}\ \texttt{add-right}$$

$$\frac{n_1\ \texttt{is a number} \qquad n_2\ \texttt{is a number}}{(\texttt{add}\ n_1\ n_2) \longrightarrow n_1 + n_2}\ \texttt{add-ok}$$

$$\frac{\dfrac{}{(+\ 1\ 2) \longrightarrow 3}\ (\text{add-ok})}{(+\ (+\ 1\ 2)\ 3) \longrightarrow (+\ 3\ 3)}\ (\text{add-left})$$

$\cancel{(+\ 3\ (+\ 1\ 2))}$

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$$\frac{e_1 \longrightarrow e_1'}{(\texttt{sub}\ e_1\ e_2) \longrightarrow (\texttt{sub}\ e_1'\ e_2)}\ \texttt{sub-left}$$

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$$\frac{n_1\ \texttt{is a number} \qquad n_2\ \texttt{is a number}}{(\texttt{sub}\ n_1\ n_2) \longrightarrow n_1 - n_2}\ \texttt{sub-ok}$$

Reduction is a Relation

$$
\begin{tikzcd}
 & (1+2)+(2+3) \arrow[dl] \arrow[dr] & \\
(1+2)+5 \arrow[dr] & & 3+(2+3) \arrow[dl] \\
 & 8 &
\end{tikzcd}
$$

It's important to recognize that **reduction is a *relation***

This means there may be multiple choices of reductions

When possible, we try do design our rules to avoid this

Reduction is a Relation

$$\frac{\text{add } 1\ 2 \longrightarrow 3}{(\text{add } (\text{add } 1\ 2)\ (\text{add } 2\ 3)) \longrightarrow (\text{add } 3\ (\text{add } 2\ 3))}\ \texttt{add-left}$$

$$\frac{\text{add } 2\ 3 \longrightarrow 5}{(\text{add } (\text{add } 1\ 2)\ (\text{add } 2\ 3)) \longrightarrow (\text{add } (\text{add } 1\ 2)\ 5)}\ \texttt{add-right}$$

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There are two reductions from $(\text{add } (\text{add } 1\ 2)\ (\text{add } 2\ 3))$ in our current rule set

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We can avoid this by *breaking symmetry*. We will enforce that the right argument can reduced only when the left argument is completely reduced

Example: Addition

```
<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
<op>   ::= add | sub | eq
<bool> ::= true | false
<int>  ::= ...
```

$$\frac{e_1 \longrightarrow e_1'}{(\texttt{add}\ e_1\ e_2) \longrightarrow (\texttt{add}\ e_1'\ e_2)}\ \texttt{add-left}$$

$$\frac{v\ \texttt{is a number} \qquad e_2 \longrightarrow e_2'}{(\texttt{add}\ v\ e_2) \longrightarrow (\texttt{add}\ v\ e_2')}\ \texttt{add-right}$$

$$\frac{n_1\ \texttt{is a number} \qquad n_2\ \texttt{is a number}}{(\texttt{add}\ n_1\ n_2) \longrightarrow n_1 + n_2}\ \texttt{add-ok}$$

(+ (+ 1 2) (+ 2 3))

Enforcing an Evaluation Order

$$\frac{\texttt{add}\ 1\ 2 \longrightarrow 3}{(\texttt{add}\ (\texttt{add}\ 1\ 2)\ (\texttt{add}\ 2\ 3)) \longrightarrow (\texttt{add}\ 3\ (\texttt{add}\ 2\ 3))}\ \texttt{add-left}$$

$$\cancel{\frac{\texttt{add}\ 2\ 3 \longrightarrow 5}{(\texttt{add}\ (\texttt{add}\ 1\ 2)\ (\texttt{add}\ 2\ 3)) \longrightarrow (\texttt{add}\ (\texttt{add}\ 1\ 2)\ 5)}}\ \texttt{add-right}$$

The new rule enforces that arguments of **add** are evaluated from left to right

Example

```
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<op>   ::= add | sub | eq
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<int>  ::= ...
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Example

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Practice Problem

```
<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
<op>   ::= add | sub | eq
<bool> ::= true | false
<int>  ::= ...
```

Write down the reduction rules for ***eq*** *(to the best of your ability) so that the left argument is evaluated before the right argument*

Answer

```
<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
<op>   ::= add | sub | eq
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<int>  ::= ...
```

ex: (num-neq)
(bool-eq)
(bool-neq)

$$\frac{e_1 \longrightarrow e_1'}{(\text{eq } e_1\ e_2) \longrightarrow (\text{eq } e_1'\ e_2)}\ (\text{eq-left})$$

$$\frac{v \text{ is a number or bool} \qquad e_2 \longrightarrow e_2'}{(\text{eq } v\ e_2) \longrightarrow (\text{eq } v\ e_2')}\ (\text{eq-right})$$

$$\frac{v_1 \text{ is num} \qquad v_2 \text{ is num} \qquad v_1 = v_2}{(\text{eq } v_1\ v_2) \longrightarrow \text{true}}\ (\text{num-eq})$$

Two Questions

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Recall: Derivations

$$\dfrac{\dfrac{\dfrac{}{(\mathsf{add}\ 1\ 2) \longrightarrow 3}\ \texttt{add-ok}}{(\mathsf{add}\ (\mathsf{add}\ 1\ 2)\ (\mathsf{add}\ 2\ 3)) \longrightarrow (\mathsf{add}\ 3\ (\mathsf{add}\ 2\ 3))}\ \texttt{add-left}}{\mathsf{sub}\ 10\ (\mathsf{add}\ (\mathsf{add}\ 1\ 2)\ (\mathsf{add}\ 2\ 3)) \longrightarrow \mathsf{sub}\ 10\ (\mathsf{add}\ 3\ (\mathsf{add}\ 2\ 3))}\ \texttt{sub-right}$$

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A **derivation** is a tree of reductions, gotten by applying reduction rules. The leaves are trivial premises

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We've done this

Recall: Building Derivations

$$\mathsf{sub}\ 10\ (\mathsf{add}\ (\mathsf{add}\ 1\ 2)\ (\mathsf{add}\ 2\ 3)) \longrightarrow \mathsf{sub}\ 10\ (\mathsf{add}\ 3\ (\mathsf{add}\ 2\ 3))$$

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» Show that $C \longrightarrow C'$

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Single-Step Evaluation

$$(\text{sub } 10\ (\text{add } (\text{add } 1\ 2)\ (\text{add } 2\ 3))) \longrightarrow ???$$

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The more "realistic" situation is to be given a program and then try to figure out what it evaluates to in a single step

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The more "realistic" situation is to be given a program and then try to figure out what it evaluates to in a single step

This is why we want to be careful about how we design our rules: *we don't want to get too caught up on which rule to apply*

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$$\frac{n \text{ is a number} \qquad e_2 \longrightarrow e_2'}{(\text{add } n\ e_2) \longrightarrow (\text{add } n\ e_2')}\ \texttt{add-right}$$

$$\frac{n_1 \text{ is a number} \qquad n_2 \text{ is a number}}{(\text{add } n_1\ n_2) \longrightarrow n_1 + n_2}\ \texttt{add-ok}$$

Example

$$(\text{sub } 10\ (\text{add } (\text{add } 1\ 2)\ (\text{add } 2\ 3))) \longrightarrow ???$$

$$\frac{\dfrac{\dfrac{}{(+\ 1\ 2) \rightarrow 3}\ (\text{add-ok})}{(+\ (+\ 1\ 2)\ (+\ 2\ 3)) \rightarrow (+\ 3\ (+\ 2\ 3))}\ (\text{add-left})}{(-\ 10\ (+\ (+\ 1\ 2)\ (+\ 2\ 3))) \rightarrow (-\ 10\ (+\ 3\ (+\ 2\ 3)))}\ \text{sub-right}$$

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$$(\texttt{sub}\ 10\ (\texttt{add}\ 3\ (\texttt{add}\ 2\ 3))) \longrightarrow (\texttt{sub}\ 10\ (\texttt{add}\ 3\ 5))$$

Give a derivation of the above reduction

Answer

$$(\text{sub } 10\ (\text{add } 3\ (\text{add } 2\ 3))) \longrightarrow (\text{sub } 10\ (\text{add } 3\ 5))$$

Multi-Step Reduction Relation

$$\frac{}{C \longrightarrow^{\star} C}\ \texttt{refl} \qquad\qquad \frac{C \longrightarrow C' \qquad C' \longrightarrow^{\star} D}{C \longrightarrow^{\star} D}\ \texttt{trans}$$

Given any single-step reduction relation, we can derive the **multi-step reduction relation:**

» Every configuration reduces to itself **(reflexivity)**

» Every $\longrightarrow^{\star}$ reduction can be extended by a single step **(transitivity)**

Two Questions (Again)

Once we have an operational semantics, there are two questions we can ask (as PL designers and on the final exam):

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How To: Derivations of Multi-Step Reductions

$$\texttt{sub}\ 10\ (\texttt{add}\ (\texttt{add}\ 1\ 2)\ (\texttt{add}\ 2\ 3)) \longrightarrow^{\star} 2$$

want to show

» Derive all necessary single-step evaluations

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$$\texttt{sub}\ 10\ (\texttt{add}\ 3\ (\texttt{add}\ 2\ 3)) \longrightarrow \texttt{sub}\ 10\ (\texttt{add}\ 3\ 5)$$ (you did this)

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$\texttt{sub}\ 10\ (\texttt{add}\ 3\ 5) \longrightarrow \texttt{sub}\ 10\ 8$ (exercise)

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$\texttt{sub}\ 10\ (\texttt{add}\ 3\ 5) \longrightarrow \texttt{sub}\ 10\ 8$ (exercise)

$\texttt{sub}\ 10\ 8 \longrightarrow 2$ (easy)

» Derive all necessary single-step evaluations

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$$\texttt{sub}\ 10\ (\texttt{add}\ (\texttt{add}\ 1\ 2)\ (\texttt{add}\ 2\ 3)) \longrightarrow^{\star} 2$$

» Derive all necessary single-step evaluations

» Combine them with the **transitivity rule**

How To: Derivations of Multi-Step Reductions

$$\dfrac{\begin{array}{c}\texttt{(we did this)}\\ \vdots\\ \mathsf{s}\ 10\ (\mathsf{a}\ (\mathsf{a}\ 1\ 2)\ (\mathsf{a}\ 2\ 3)) \longrightarrow \mathsf{s}\ 10\ (\mathsf{a}\ 3\ (\mathsf{a}\ 2\ 3))\end{array} \qquad \mathsf{s}\ 10\ (\mathsf{a}\ 3\ (\mathsf{a}\ 2\ 3)) \longrightarrow^{\star} 2}{\mathsf{sub}\ 10\ (\mathsf{add}\ (\mathsf{add}\ 1\ 2)\ (\mathsf{add}\ 2\ 3)) \longrightarrow^{\star} 2}\ \texttt{trans}$$

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If our rules are well defined, then should be easy:

Solve this single-step evaluation problem until you reach a configuration that cannot be further reduced

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want to show

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`y (fun x -> x)` $\not\rightarrow$

» **diverge:** the computation never reaches a point where the expression is not reducible

`(fun x -> x x) (fun x -> x x)` $\rightarrow^{\star}$
`(fun x -> x x) (fun x -> x x)`

moving onto big-step...

Big-Step Semantics

$$(\mathsf{sub}\ 10\ (\mathsf{add}\ (\mathsf{add}\ 1\ 2)\ (\mathsf{add}\ 2\ 3))) \Downarrow 2$$

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Notation: We write $e \Downarrow v$ to mean that e evaluates to the value v

This is what we've been doing in this course so far

Example

```
<expr> ::= ( <op> <expr> <expr> )
         | <bool> | <int>
<op>   ::= add | sub | eq
<bool> ::= true | false
<int>  ::= ...
```

$$\frac{n \text{ is a number}}{n \Downarrow n}\ \texttt{numEval}$$

$$\frac{e_1 \Downarrow v_1 \qquad e_2 \Downarrow v_2 \qquad v_1 \text{ is a number} \qquad v_2 \text{ is a number}}{(\texttt{add}\ e_1\ e_2) \Downarrow v_1 + v_2}\ \texttt{addEval}$$

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we'll remove these side conditions once we have type-checking

Practice Problem

```
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         | <bool> | <int>
<op>   ::= add | sub | eq
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```

Write the rule for ***eq***

Answer

```
<expr> ::= ( <op> <expr> <expr> )
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Note: We can't always have both small-step and big-step!

Order of Evaluation

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We will take the order of operations to be from left to right

Summary

big-step

$$e \Downarrow v$$

e evaluates to v

single-step

$$e \longrightarrow e'$$

e reduces to e' in a single step

multi-step

$$e \longrightarrow^{\star} e'$$

e reduces to e' in many steps